



BUSINESS REPORT

Python For Data Science: Austo Automobiles Case Study



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1. Introduction

Austo Motor Company is a leading car manufacturer specializing in SUV, Sedan, and Hatchback models. In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used. The board decides to rope in an analytics professional to improve the existing campaign.

2. Objective

They want to analyze the data to get a fair idea about the demand of customers which will help them in enhancing their customer experience. Suppose you are a Data Scientist at the company and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company to improve the business.

3. Data Description

- Age: The age of the individual in years.
- Gender: The gender of the individual, categorized as male or female.
- Profession: The occupation or profession of the individual.
- Marital_status: The marital status of the individual, such as married & single
- Education: The educational qualification of the individual Graduate and Post Graduate
- No_of_Dependents: The number of dependents (e.g., children, elderly parents) that the individual supports financially.
- Personal_loan: A binary variable indicating whether the individual has taken a personal loan "Yes" or "No"
- House_loan: A binary variable indicating whether the individual has taken a housing loan "Yes" or "No"
- Partner_working: A binary variable indicating whether the individual's partner is employed "Yes" or "No"
- Salary: The individual's salary or income.
- Partner_salary: The salary or income of the individual's partner, if applicable.
- Total_salary: The total combined salary of the individual and their partner (if applicable).
- Price: The price of a product or service.
- Make: The type of automobile.

4. Data Collection & Preprocessing:

- *Imported all the libraries*

```
[ ] # importing all the libraries
import pandas as pd
import numpy as np

# libraries to help with data visualization
import matplotlib.pyplot as plt
import seaborn as sns

# to restrict the float value to 3 decimal places
pd.set_option('display.float_format', lambda x: '%.3f' % x)

# Command to tell Python to actually display the graphs
%matplotlib inline
```

- **Loaded the dataset connecting to Google Drive and imported**

```
[ ] # Connect to Drive
from google.colab import drive
drive.mount('/content/drive')
```

 Mounted at /content/drive

```
[ ] data = pd.read_csv ('/content/drive/MyDrive/Colab Notebooks/austo_automobile .csv')
```

```
[ ] # making a copy of the dataset
df = data.copy()
```

- **Data overview by performing simple steps**

1. Observing the first few rows of the dataset

```
[ ] data.head()
```



	Age	Gender	Profession	Marital_status	Education	No_of_Dependents	Personal_loan	House_loan	Partner_working	Salary	Partner_salary	Total_salary	Price	Make
0	53	Male	Business	Married	Post Graduate	4	No	No	Yes	99300	70700.000	170000	61000	SUV
1	53	Femal	Salaried	Married	Post Graduate	4	Yes	No	Yes	95500	70300.000	165800	61000	SUV
2	53	Female	Salaried	Married	Post Graduate	3	No	No	Yes	97300	60700.000	158000	57000	SUV
3	53	Female	Salaried	Married	Graduate	2	Yes	No	Yes	72500	70300.000	142800	61000	SUV
4	53	Male	Salaried	Married	Post Graduate	3	No	No	Yes	79700	60200.000	139900	57000	SUV

2. Observing the last few rows of the dataset

```
[ ] data.tail()
```



	Age	Gender	Profession	Marital_status	Education	No_of_Dependents	Personal_loan	House_loan	Partner_working	Salary	Partner_salary	Total_salary	Price	Make
1576	22	Male	Salaried	Single	Graduate	2	No	Yes	No	33300	0.000	33300	27000	Hatchback
1577	22	Male	Business	Married	Graduate	4	No	No	No	32000	NaN	32000	31000	Hatchback
1578	22	Male	Business	Single	Graduate	2	No	Yes	No	32900	0.000	32900	30000	Hatchback
1579	22	Male	Business	Married	Graduate	3	Yes	Yes	No	32200	NaN	32200	24000	Hatchback
1580	22	Male	Salaried	Married	Graduate	4	No	No	No	31600	0.000	31600	31000	Hatchback

3. Getting information about the number of rows and columns.

```
[ ] data.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1581 entries, 0 to 1580  
Data columns (total 14 columns):  
#   Column              Non-Null Count  Dtype  
---  ---  
0   Age                 1581 non-null  int64  
1   Gender              1528 non-null  object  
2   Profession          1581 non-null  object  
3   Marital_status      1581 non-null  object  
4   Education           1581 non-null  object  
5   No_of_Dependents    1581 non-null  int64  
6   Personal_loan       1581 non-null  object  
7   House_loan          1581 non-null  object  
8   Partner_working     1581 non-null  object  
9   Salary              1581 non-null  int64  
10  Partner_salary      1475 non-null  float64  
11  Total_salary        1581 non-null  int64  
12  Price               1581 non-null  int64  
13  Make                1581 non-null  object  
dtypes: float64(1), int64(5), object(8)  
memory usage: 173.1+ KB
```

Findings from the dataset:

- There are total 6 columns which are numerical and rest are categorical in nature (Float, Int in data type).
- Observing the data, seems there are missing values in 'Gender' and 'Partner Salary'.
- There are total 1581 rows and 14 columns in the dataset.
- There are 0 duplicates in the dataset.

5. Checking the statistical summary

```
[ ] # Checking for statistical summary of the data  
data.describe()
```

```
<class 'pandas.core.frame.DataFrame'>  
# Checking for statistical summary of the data  
data.describe()
```

	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	Price
count	1581.000	1581.000	1581.000	1475.000	1581.000	1581.000
mean	31.922	2.458	60392.220	20225.559	79625.996	35597.723
std	8.426	0.943	14674.825	19573.149	25545.858	13633.637
min	22.000	0.000	30000.000	0.000	30000.000	18000.000
25%	25.000	2.000	51900.000	0.000	60500.000	25000.000
50%	29.000	2.000	59500.000	25600.000	78000.000	31000.000
75%	38.000	3.000	71800.000	38300.000	95900.000	47000.000
max	54.000	4.000	99300.000	80500.000	171000.000	70000.000

- The age of customers ranges between minimum 22 to maximum 54, Average age of a vehicle-buying customer being approximately 31.
- Customers typically have a maximum of four dependents.
- The Total salary vary between minimum \$30,000 to maximum \$1,71,000 and the average salary is approximately \$80,000.
- As per the data Vehicle prices range from a minimum of 18,000 dollars to a maximum of \$70,000, average price of the vehicle is around \$36,000.

Let's check and treat missing values from the dataset:

```

0s [✓] # checking missing values from the dataset
data.isnull().sum()

```

	0
Age	0
Gender	53
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	106
Total_salary	0
Price	0
Make	0

dtype: int64

We have observed the missing values in Gender and Partner Salary column

Missing values in Gender – 53

Missing values in Partner Salary 106

For the gender field we have observed that it does not have any relation with any other field hence we have treated the gender missing values using Mode function which was highest

frequency in the data (added male category in the missing values as it was the highest frequency)

While for missing values in Partner salary we observed that it is highly depend on Working condition of the partner, Total salary and and salary.

Where partner working condition is yes we have treated the date using formula as

Partner_Salary = Total_Salary – Salary

Also where the partners working condition is mentioned as No we assigned 0 salary for the same.

6. Checking for anomalies in the data.

```
✓ [0s] # checking the gender column as there were some wrong values in the data
data['Gender'].unique()

array(['Male', 'Femal', 'Female', nan, 'Femle'], dtype=object)
```

There are a total of 4 values in gender, two of which are incorrect

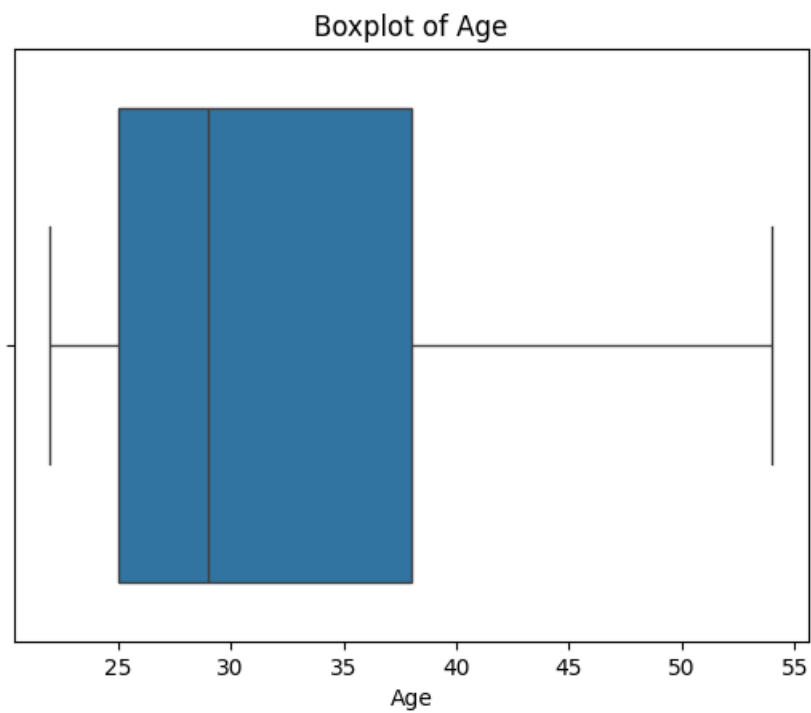
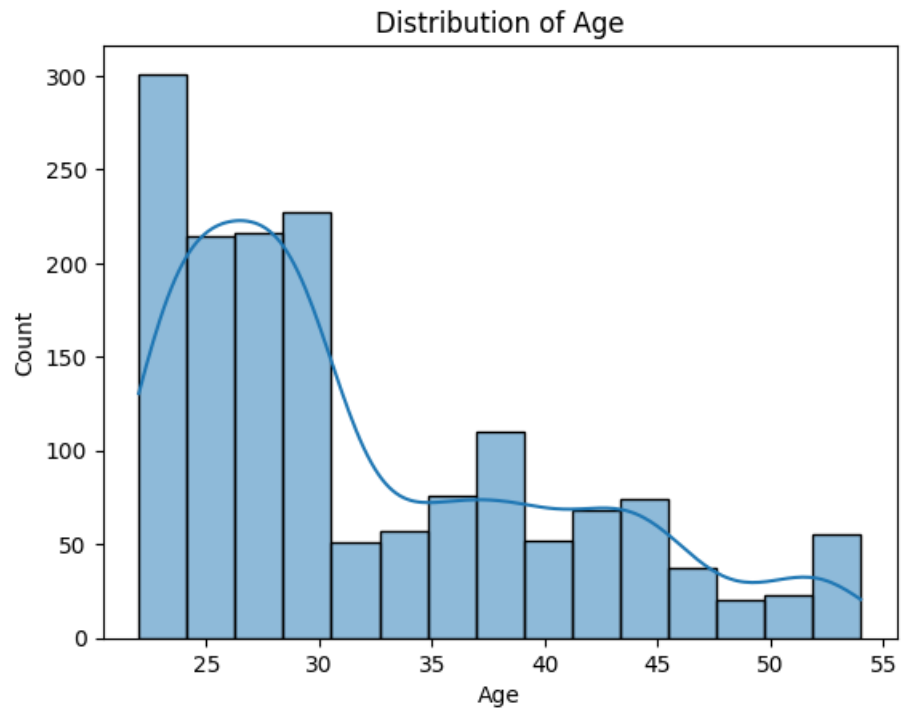
```
✓ [73s] data['Gender'].value_counts()

count
Gender
Male    1199
Female   327
Femal     1
Femle     1
dtype: int64
```

We have replaced the same by using replace function

7. Univariate analysis of numeric variables:

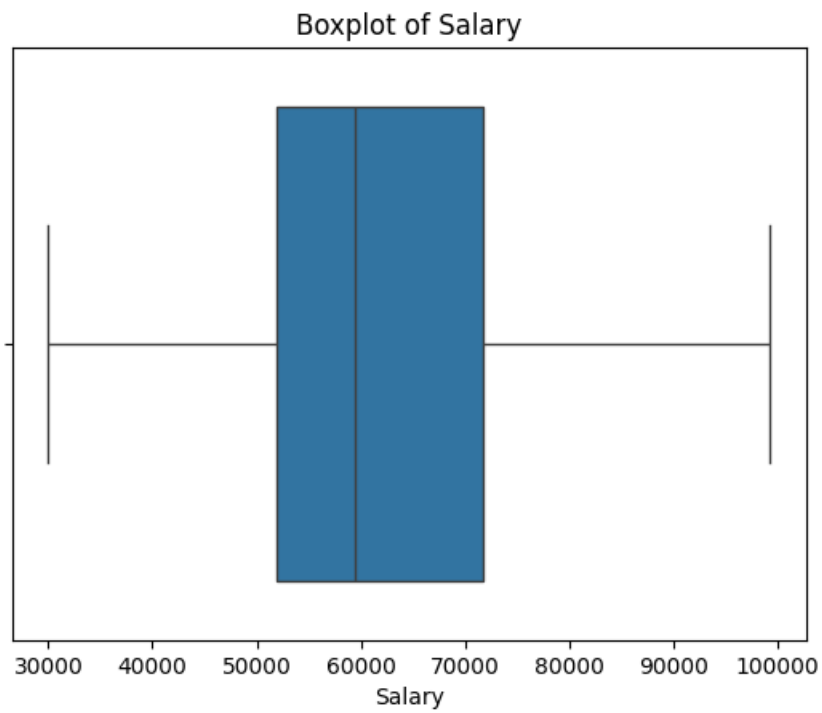
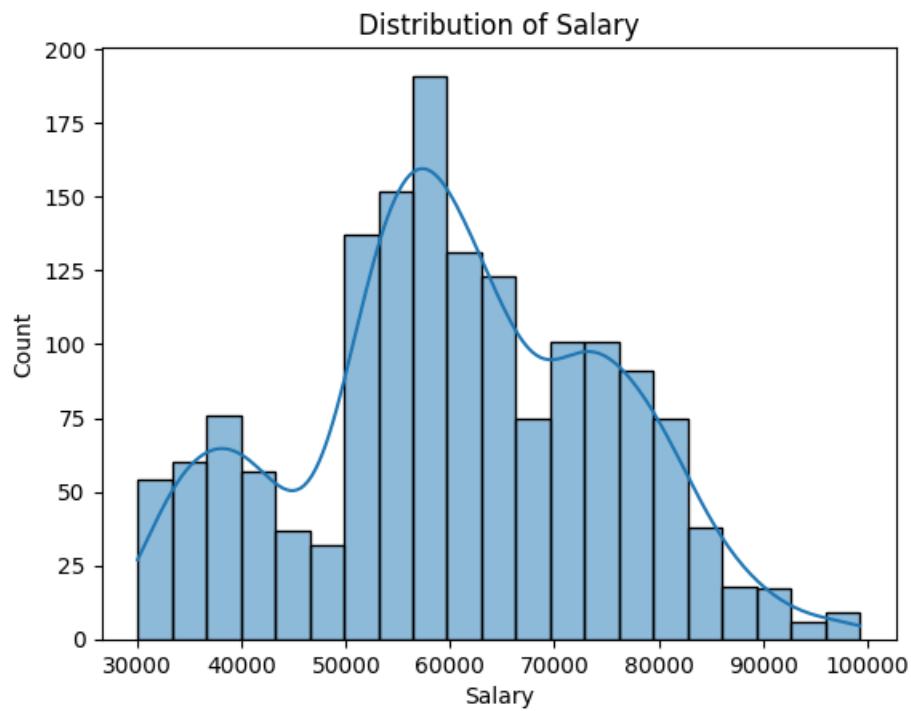
Univariate analysis of Age-



Observations:

- The distribution is skewed to the right with no outliers.
- The median age is around 29 and the maximum age is 54.

Univariate analysis of Salary-

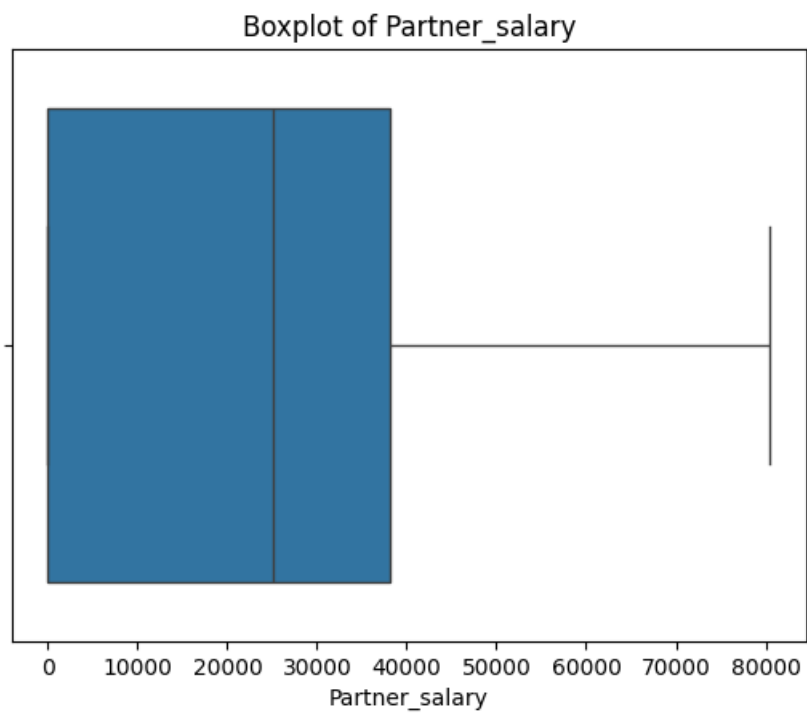
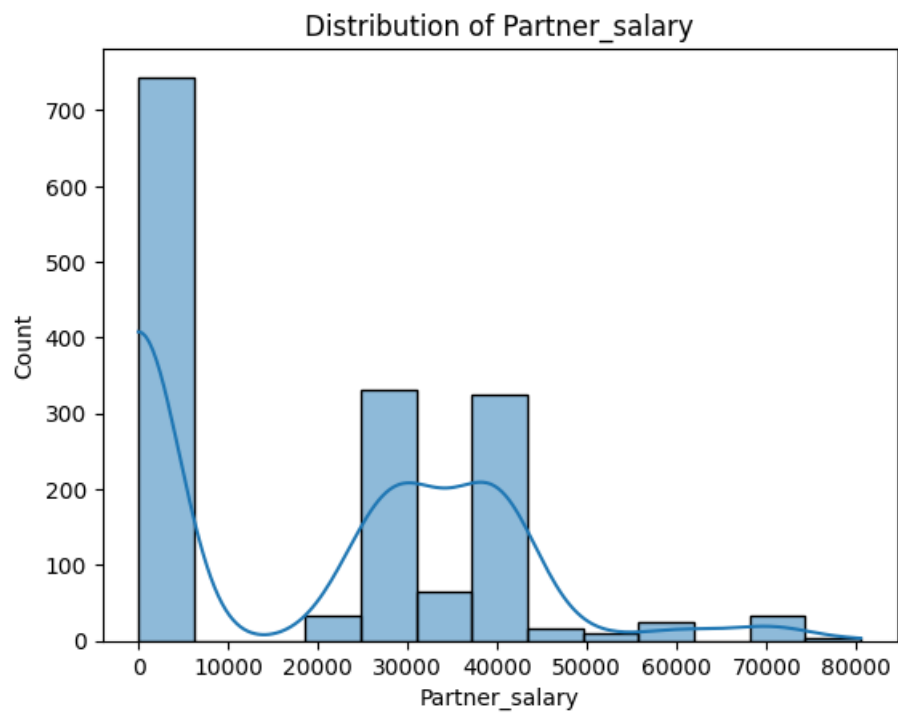


Observation

- *The distribution is balanced with no outliers.*

- The distribution is normal and median salary is around \$59000 and the maximum salary is around \$99000

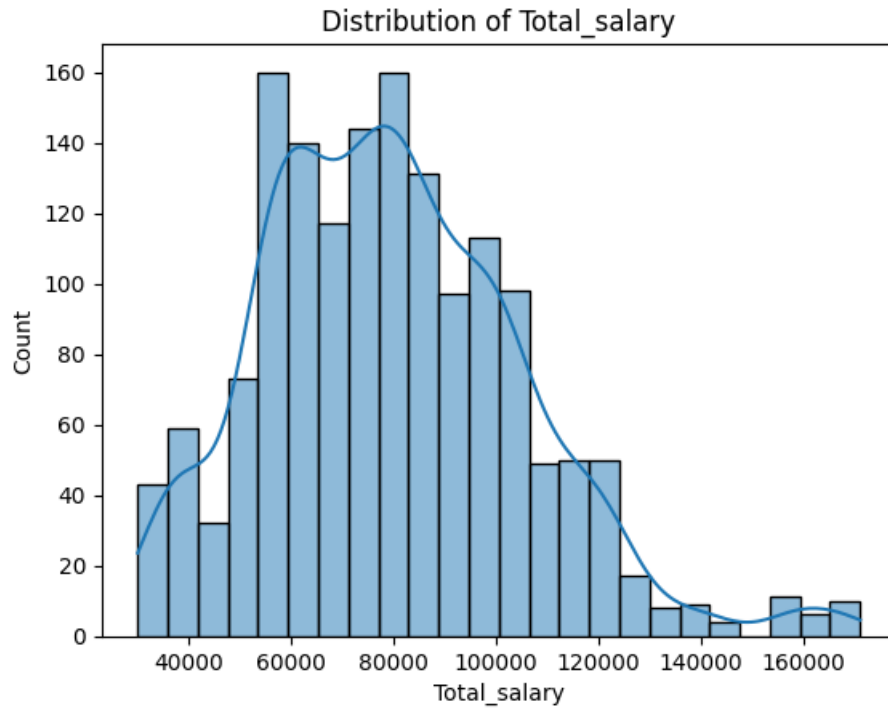
Univariate analysis of Partners Salary

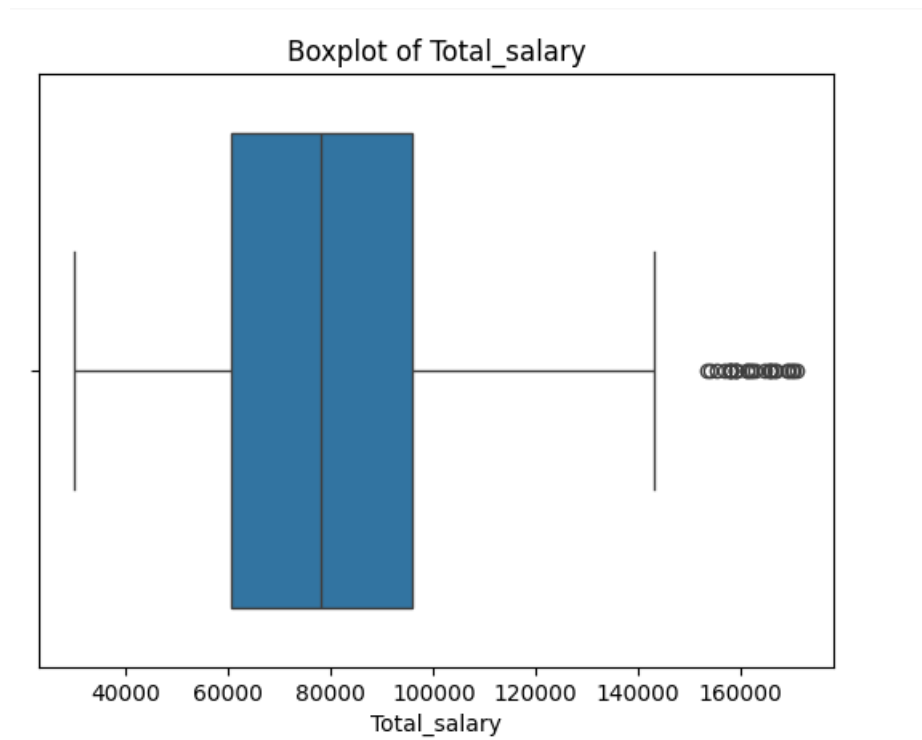


Observations

- *The distribution is skewed to the right with no outliers.*
- *The median salary is around \$25000 and the maximum total salary is \$80000*

Univariate analysis of Total_salary

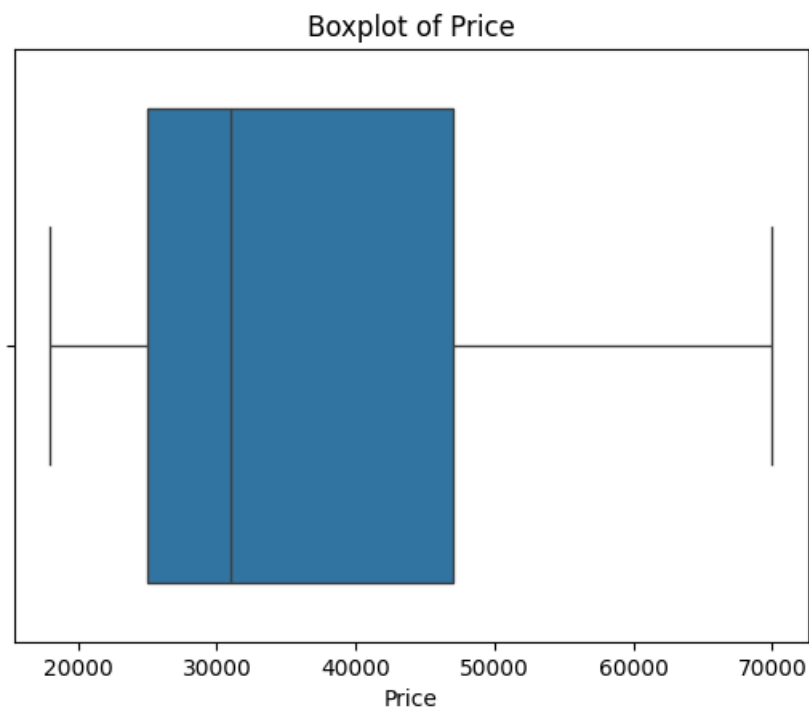
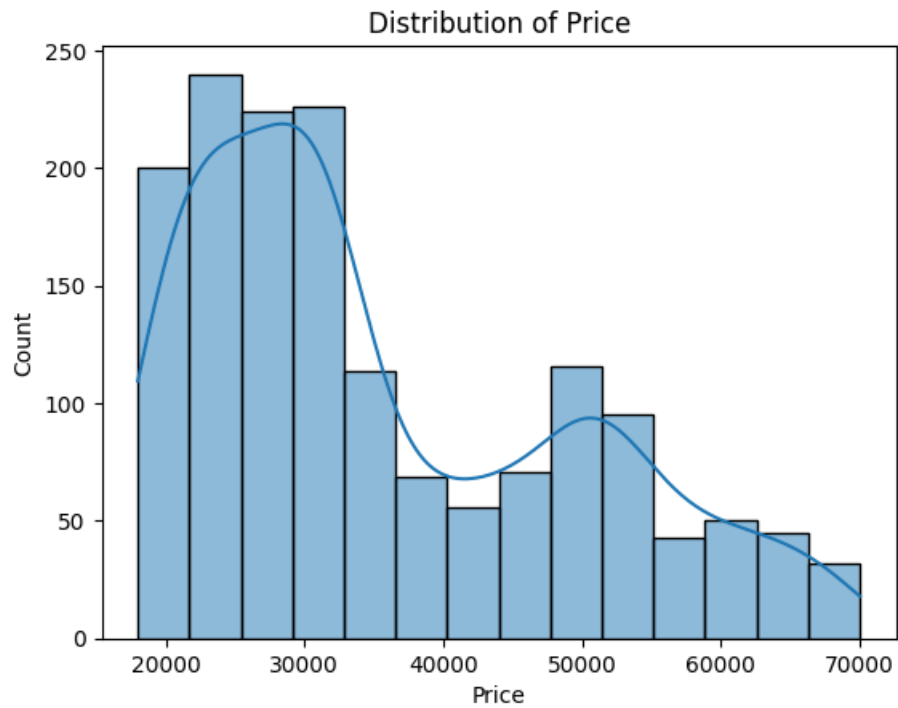




Observation

- *The distribution is slightly skewed to the right and has some outliers.*
- *It's been observed that the total salary is highly impacted due to individual salary and partner's salary.*
- *The median total salary is around \$78000.*

Univariate analysis of Price



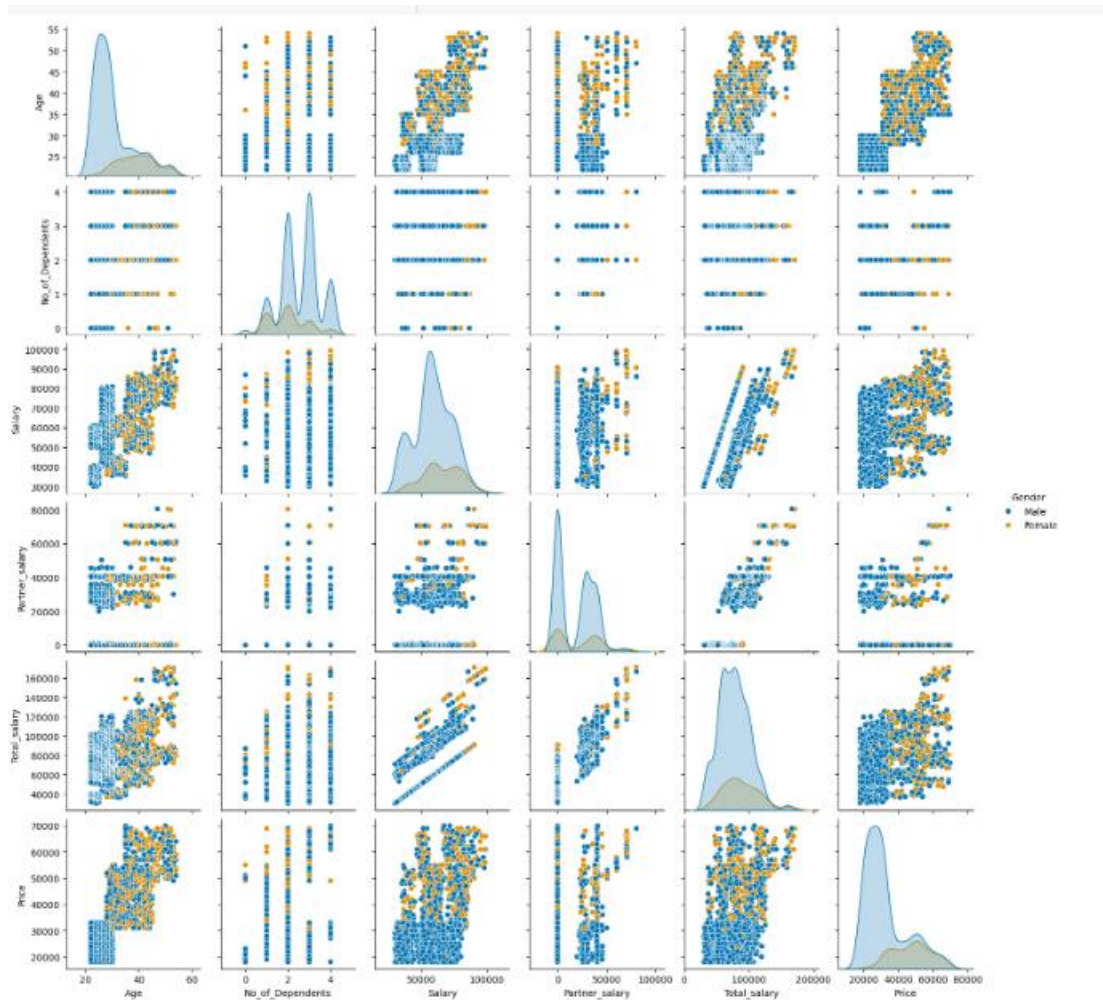
Observation on Price

- *The distribution is skewed to the right with no outliers.*

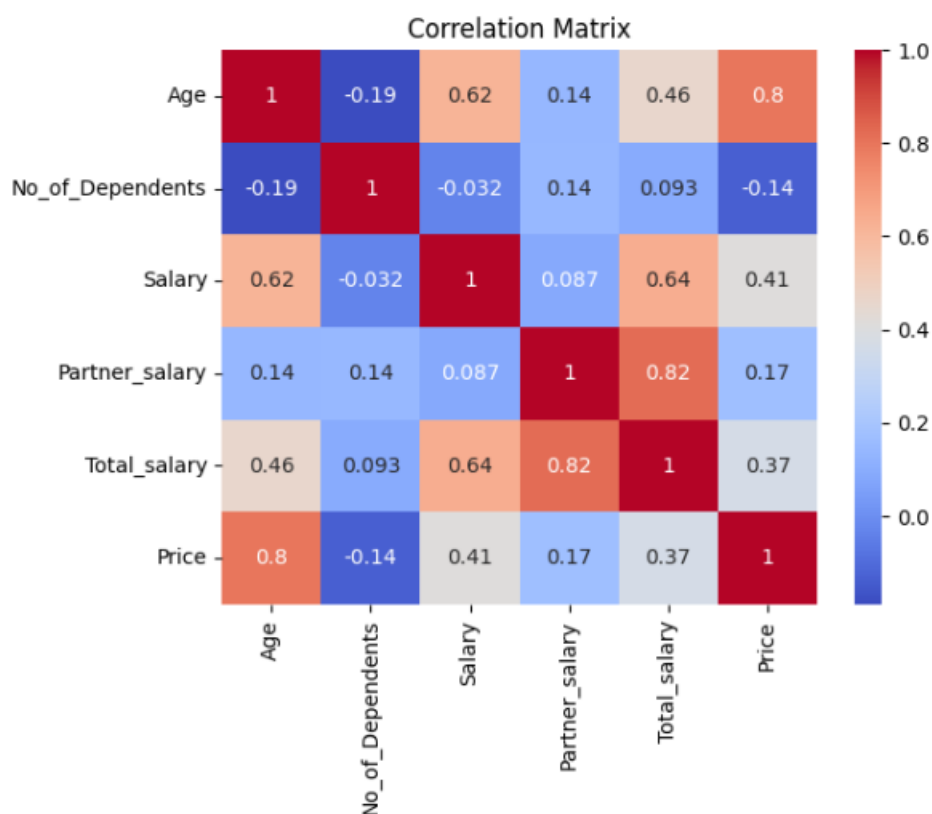
- The median price is around 32000 dollars, and the maximum price is 70000 dollars.

- **Bivariate Analysis**

- Let's try to visualize how these numerical variables are associated with each other



	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	Price
Age	1.000	-0.190	0.617	0.136	0.459	0.798
No_of_Dependents	-0.190	1.000	-0.032	0.144	0.093	-0.136
Salary	0.617	-0.032	1.000	0.087	0.642	0.410
Partner_salary	0.136	0.144	0.087	1.000	0.820	0.172
Total_salary	0.459	0.093	0.642	0.820	1.000	0.368
Price	0.798	-0.136	0.410	0.172	0.368	1.000

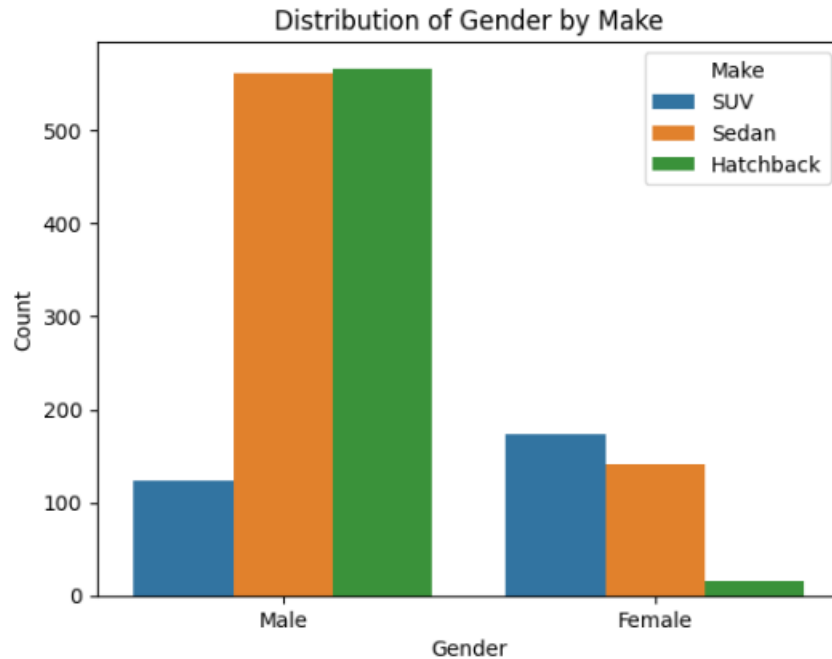


Observation

- The price and Age is highly positively correlated with the value 0.8.
- Also secondly, TotSalary and Partner_Salary are positively correlated with each other.
- No_of_dependents is negatively correlated with price.

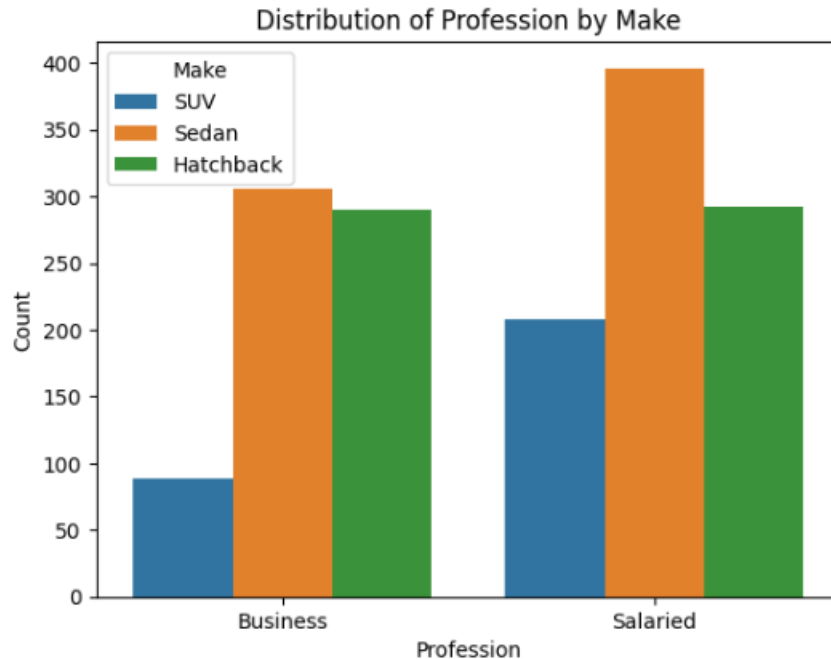
Let's explore the data to answer the following key questions:

- **Do men tend to prefer SUVs more compared to women?**



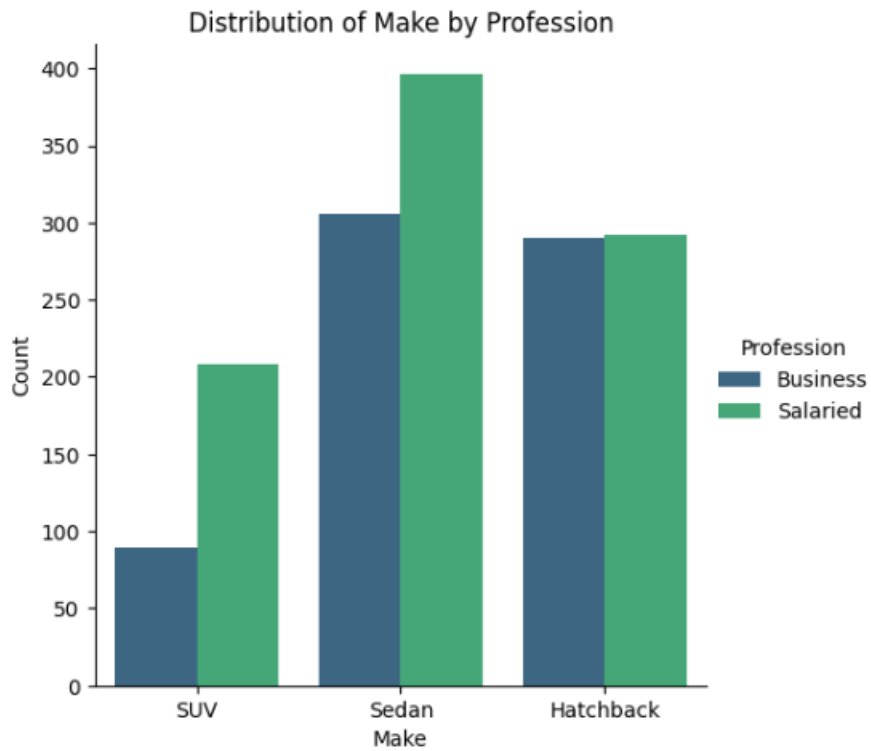
Observation

- *Men do not show a preference for SUVs, Its been observed that males are more tend to buy Hatchbacks, followed by sedan.*
 - *Women are purchasing the highest number of SUVs at 173 followed by sedan.*
 - *The breakdown of SUV buyers shows 58% are female and 41% are male.*
-
- **What is the likelihood of a salaried person buying a Sedan?**



Observation

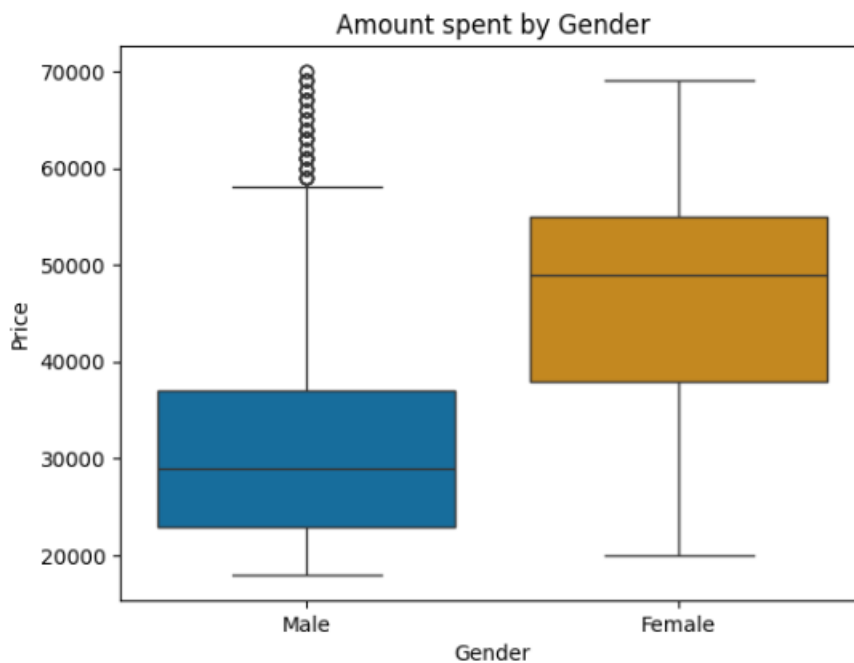
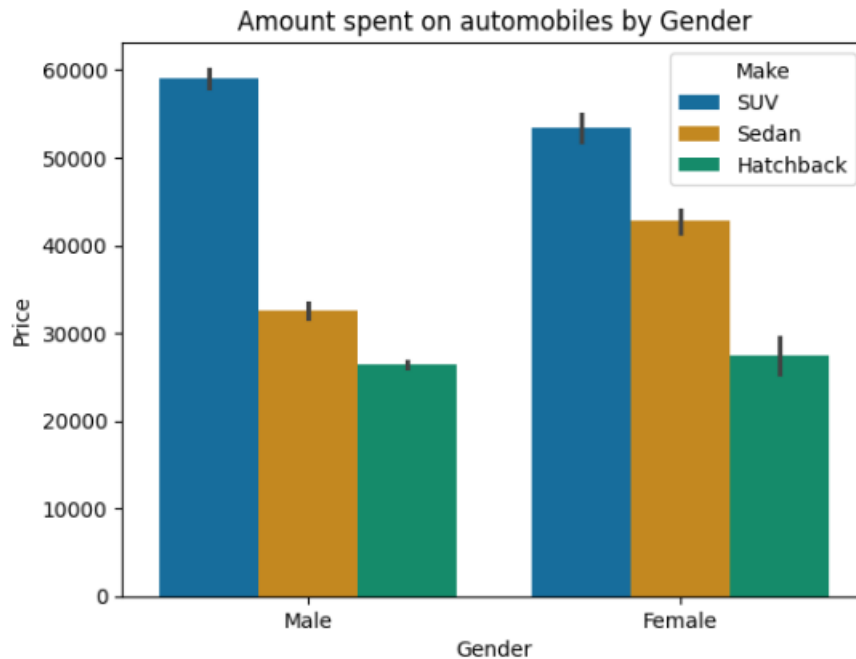
- *The salaried professionals prefer to buy sedan as compare to business professionals with the percentage of 56%.*
 - *Also Hatchback is equally distributed among both the professionals.*
 - *Among buyers, salaried professionals constitute 23% of SUV purchasers, 32% of hatchback purchasers, and 44% of sedan purchasers.*
-
- **What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?**



Observation

- *Our observations indicate that salaried professionals demonstrate a higher propensity to purchase sedans, with hatchbacks being their second preference. They show a lower likelihood of preferring SUVs.*

4. How does the amount spent on purchasing automobiles vary by gender?



Observation

- Females tend to spend significantly more on automobiles than men, with an observed average of 134% higher spending.
- Observed few outliers on male spending on automobiles.
- The median amount spent is \$28,000 for males and \$48,000 for females.
- Females tend to spend more on SUVs, followed by sedans.

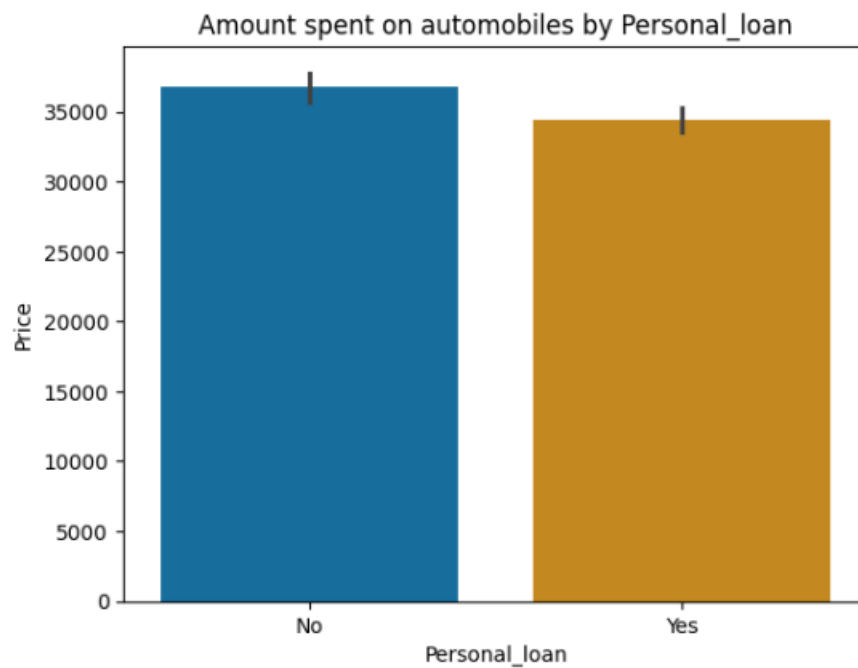
- *Males are more inclined to buy SUVs than females.*

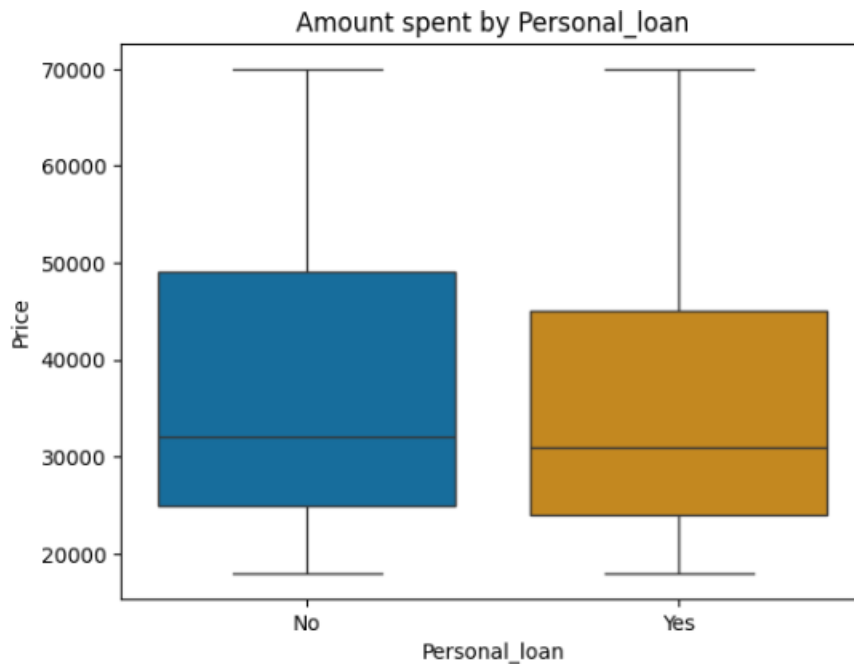
5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

	count	mean	std	min	25%	50%	75%	max
Personal_loan								
No	789.000	36742.712	14534.345	18000.000	25000.000	32000.000	49000.000	70000.000
Yes	792.000	34457.071	12578.780	18000.000	24000.000	31000.000	45000.000	70000.000

Mean amount spent by an individual who took a personal loan = \$34457.07

Mean amount spent by individuals who have not taken a personal loan = \$36742.71





Observation

- It's been highly observed that there is not much difference to be seen in the purchasing limit based on whether the individual who taken a personal loan or not.
- The numbers are slightly high for the individual who did not take any personal loan.
- The average amount spent is around \$37000 who did not take any personal loan; however, it is around \$15000 who took the loan.
- There are no outliers.

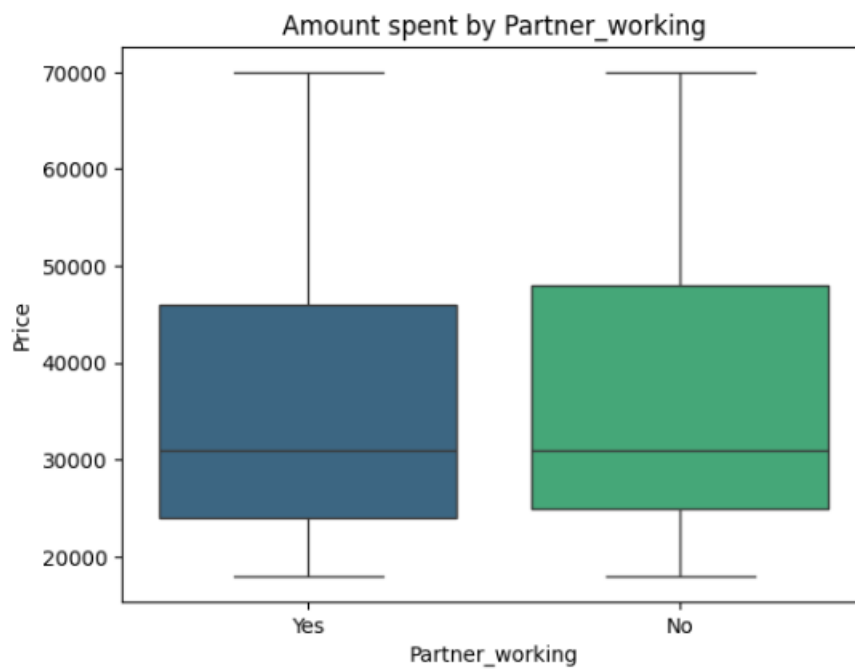
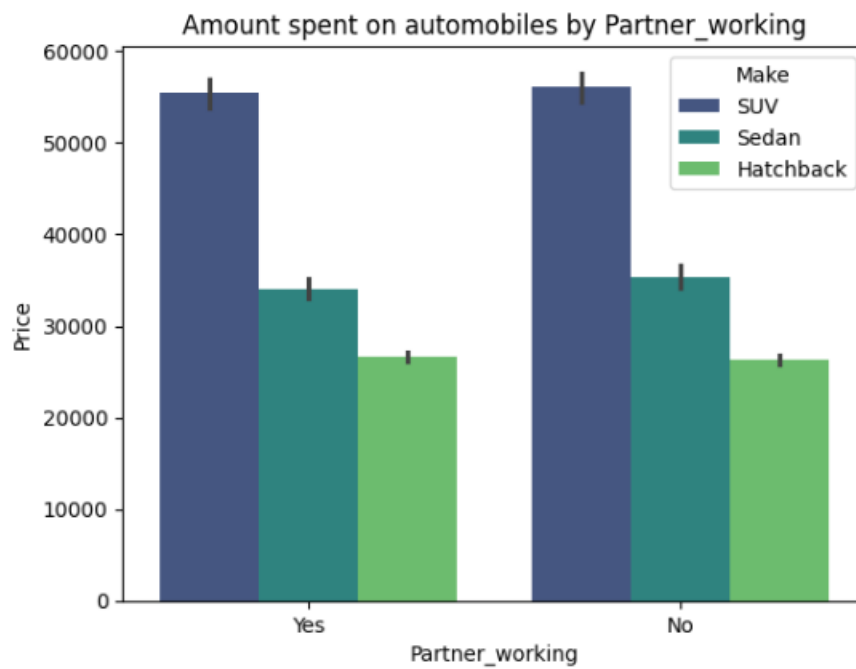
6. How does having a working partner influence the purchase of higher-priced cars?

	count	mean	std	min	25%	50%	75%	max
Partner_working								
No	713.000	36000.000	13817.734	18000.000	25000.000	31000.000	48000.000	70000.000
Yes	868.000	35267.281	13479.533	18000.000	24000.000	31000.000	46000.000	70000.000

Average amount spent by partner not working = \$36000

Average amount spent by working partner = \$35267.53

Median is similar for both = \$31000



Observation

- It's been highly observed that Partner working condition does not have much impact on purchase of high priced cars.

- *There are no outliers.*

Actionable Insights and Recommendations from the data

1. Do men tend to prefer SUVs more compared to women?

Actionable Insight-

- We have identified that Males did not show a preference to buy SUVs however they are more tend to buy Hatchbacks followed by Sedan it is suggested that marketing efforts and sales strategies specifically targeting male consumers might be misaligned if they heavily emphasize SUVs. There's a clear opportunity to reallocate marketing budgets and tailor promotional content to spotlight the benefits of Hatchbacks and Sedans

Recommendations -

- Tailor promotional content (focusing on specific benefits of Hatchbacks/Sedans) and for women (On SUVs)

2. What is the likelihood of a salaried person buying a Sedan?

Actionable Insight-

Our analysis reveals that out of 56% of the salaried individuals preferred to buy sedans than business professionals This indicates that our current sedan marketing and product positioning effectively resonate with the lifestyle, commuting needs, and budget considerations of salaried individuals, presenting a prime opportunity to double down on marketing efforts for sedans specifically tailored to this high-conversion segment.

Recommendations -

- Develop and launch targeted digital and traditional marketing campaigns (e.g., social media ads, email marketing, print ads in relevant publications/websites) specifically highlighting sedan features that appeal most to salaried individuals.
- Conduct training for sales teams to educate them on the distinct preferences and common pain points of salaried professionals when considering car purchases.
- Explore partnerships with large corporations, professional associations, or public sector organizations that employ a high number of salaried professionals. Offer exclusive corporate discounts, preferred financing rates, or special service packages on sedans for their employees. Participate in corporate events, job fairs, or online forums popular among this group.

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

Actionable insight-

- Our analysis shows that salaried professionals prioritize sedans, followed by hatchbacks. They show a lower likelihood of preferring SUVs, it is suggested that reallocate resources and intensify marketing for sedans and hatchbacks, specifically targeting salaried professionals.

Recommendations-

- Reallocate Marketing Budget & Focus to Sedans and Hatchbacks, targeting salaried professionals.
- Develop Tailored Messaging & Content for Salaried Lifestyles by creating marketing collateral, ad copy, and social media content that directly speaks to the daily life and aspirations of salaried individuals.
- We can also Optimize Inventory and Showroom Presentation by ensuring that dealerships maintain robust inventory levels of popular sedan and hatchback models and variants that appeal to salaried buyers.

4. How does the amount spent on purchasing automobiles vary by gender?

Actionable insight-

- Our analysis reveals that women represent an exceptionally high-value segment in automobile purchases, spending an average of 134% more than men. It is suggested to re-evaluate our offerings, messaging, and sales approach to better align with the factors driving this higher spending by women.

Recommendations-

- Develop targeted and Empowering Marketing Campaigns for female buyers.
- We can also Tailor Product Offerings and Showroom Experience to Female Preferences ensuring that vehicle with safety and specific body types (like SUVs, as previously identified) that contribute to higher spending are prominently featured in marketing and readily available in inventory.

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

Actionable Insight-

- Individuals who took a personal loan tend to spend average amount which is \$36000 dollars and maximum of \$45000 on automobiles.

Recommendations-

- We can optimize Inventory and Promotions for the \$35,000 - \$45,000 Price Segment.
- We can also collaborate with financing partners to offer highly competitive auto loan rates, flexible payment structures, or perhaps even deferred payment options specifically designed for customers who may already be managing personal loan debt.

6. How does having a working partner influence the purchase of higher-priced cars?

Actionable Insight-

- Analysis reveals that the presence of a working partner has minimal impact on the likelihood of purchasing high-priced automobiles.

Recommendations-

- We can consider prioritizing high value customer segments which is profession and gender as partner working condition is not a strong predictor for identifying potential high-priced car buyers.

Conclusion-

The data advocates for a highly segmented and adaptive marketing and sales strategy. By redirecting resources based on actual consumer behavior – celebrating women's higher spending and SUV preference, catering to salaried professionals' sedan affinity, understanding loan-takers' spending patterns, and de-emphasizing partner employment as a high-value predictor – the business can significantly enhance its market penetration, conversion rates, and overall profitability.